Eikonal Bridge Quick Reference

Differentiable Eikonal Engine & Classical–Quantum Bridge

The Eikonal Bridge: From Classical Lens Design to Quantum Photonics

Book 1 of the Open-Source Engineering Optics Trilogy

Version 4.2 | Standalone practitioner reference

What To Do First

Standalone desk reference for the Differentiable Eikonal Engine methodology. For full derivations see Chapters 1–8. Work through the embedded WE-7.0 condensed example (§12) before tackling real systems.
New in v4.2: Genesis section (§1); DR-E.11 "The Unity Rule"; classical–quantum failure coupling table (Table 13); completed WE-7.0 quantum re-optimization; expanded JAX starter (§22); thermal path-length clarification; Walther & Matsui-Nariai in annotated references.

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1 Genesis: The Question That Started It All

§0 — "Why are quantum photonics tolerances so impossibly tight?"

The standard answer — quantum states are fragile, phase matters more, etc. Could we give a *number*? Even just an imprecise number?
That question, rooted over two decades of teaching and practice (Walther's *The Ray and Wave Theory of Lenses* since ∼2002; Matsui & Nariai's *Fundamentals of Practical Aberration Theory* since ∼2005), led to what eventually became the **bridge identity**:

$$\boxed{\phi_{\text{quantum}} = \frac{2\pi}{\lambda}\, W_{\text{eikonal}}} \tag{BI}$$

The answer turns out to be something worthwhile to note.
Classical Strehl ratio S and quantum fidelity F are the **same number** — literally the same overlap integral:

$$S = \left| \iint P^*_{\text{ideal}}\, P_{\text{actual}}\, d^2\mathbf{r} \right|^2 = F \tag{1}$$

The Maréchal criterion ($\sigma_W < \lambda/14$) gives $S > 0.80$. Quantum photonics ($F > 0.99$) requires $\sigma_W < \lambda/63$. That is a **4.7×** tightening factor.
For quantum computing gates ($F > 0.9999$)? The factor reaches **45×**.
Once you see this, the entire classical-to-quantum translation becomes *systematic*. Surface roughness: $2\,\text{nm} \to 0.2\,\text{nm}$ (10×). Alignment: $10\,\mu\text{rad} \to 0.1\,\mu\text{rad}$ (100×). Temperature stability: $1\,\text{K} \to 10\,\text{mK}$ (100×).

The punchline: The same JAX code that optimizes your camera lens also optimizes a quantum photonic gate. You write `forward_model()` once. Classical uses it directly. Quantum wraps it in a different loss function. *One equation. Two domains. Unified code.*

2 The Eikonal Bridge Paradigm

Any optical system—classical or quantum—can be described through the eikonal function $W(\mathbf{r};\mathbf{p})$, which satisfies:

$$|\nabla W|^2 = n^2(\mathbf{r}) \tag{2}$$

The eikonal W is the optical path length (OPL) from source to field point. It simultaneously encodes ray direction ($\nabla W = n\,\hat{s}$), wavefront shape ($W = \text{const}$ surfaces), and—via the bridge identity—quantum phase.

2.1 The Bridge Identity

$$\phi_{\text{quantum}} = \frac{2\pi}{\lambda}\, W_{\text{eikonal}} \tag{3}$$

This is an exact mathematical identity (from the WKB limit), not merely an analogy. It means: the same code, the same Zernikes, the same tolerancing machinery work in both domains. Only the loss function target changes.

2.2 The Maréchal Approximation

The Strehl ratio for small wavefront errors is:

$$S \approx e^{-(2\pi\sigma_W)^2} \tag{4}$$

where σ_W is the RMS wavefront error in waves. Via the bridge identity, $\sigma_\phi = 2\pi\sigma_W/\lambda$ and $F = S = e^{-\sigma_\phi^2}$.

2.3 Units and Scaling Sanity Check

Table 1: Typical parameter ranges for quick order-of-magnitude sanity checks.

Parameter	Typical range	Unit	Example
σ_W (consumer)	0.07–0.15	waves	$S = 0.5$–0.8
σ_W (precision)	0.02–0.07	waves	$S > 0.8$
σ_W (quantum)	0.005–0.016	waves	$F > 0.99$
λ (visible)	400–700	nm	He-Ne: 632.8 nm
λ (telecom)	1310–1550	nm	C-band: 1550 nm
NA (imaging)	0.05–0.30	—	Camera lens: 0.12
NA (microscopy)	0.5–1.4	—	Oil immersion: 1.35
n (glass)	1.45–1.95	—	BK7: 1.5168; SF11: 1.785
Surface roughness	0.2–5	nm RMS	IBF: 0.2; std polish: 2

Quick check: For a singlet with $\sigma_W = 0.784$ waves at $\lambda = 550$ nm, the Strehl ratio is $S = e^{-(2\pi\times 0.784)^2} = 0.005$ (terrible). The quantum phase error is $\sigma_\phi = 2\pi \times 0.784 = 4.93$ rad, yielding fidelity $F = e^{-\sigma_\phi^2} \approx 0$ (state destroyed). Both S and F measure the same overlap integral.

2.4 The Walther–(Matsui-Nariai) Duality

Every optical engineering problem is either:

- **WALTHER (Forward Analysis):** Given device → predict performance.
  Input: System parameters $\{p_j\}$ → `forward_model()` → Output: S, MTF, F.
  "What does this lens do?"
- **MATSUI-NARIAI (Inverse Design):** Given target → design device.
  Input: Target spec → `loss()` + `jax.grad` → Output: Optimized $\{p_j^*\}$.
  "How do I build it?"

The same `dee_forward_model()` serves both—Walther uses it directly; Matsui-Nariai wraps it in a loss function and differentiates.

Design Rule DR-E.1 — Bridge Classification

For every optical problem, classify: is this Walther (analysis) or Matsui-Nariai (design)? The DEE core is shared. Only the wrapper differs: direct call (W) or `jax.grad(loss)` (MN).

2.5 Classical–Quantum Correspondence Table

Table 2: Classical ⟷ Quantum correspondence via bridge identity.

Classical	Bridge	Quantum
Wavefront $W(x,y)$ [waves]	$\phi = 2\pi W/\lambda$	Phase $\phi(x,y)$ [rad]
Zernike coefficients $\{a_n\}$	Equal values	Quantum observables $\{a_n\}$
RMS WFE σ_W	$\Delta\phi = 2\pi\sigma_W/\lambda$	Phase uncertainty $\Delta\phi$
Strehl ratio S	$F = S$	State fidelity F
Maréchal: $\sigma_W < \lambda/14$	—	$F > 0.80$ (classical limit)
PSF	—	Wigner function
MTF	—	Entanglement witness
Hessian H_{ij}	$F_Q = 4H$	Quantum Fisher Information

3 Level Selection: The Five-Level Hierarchy

The eikonal equation (2) is Level 1—the "workhorse" model. When it fails, a five-level hierarchy provides systematic upgrades.

Table 3: The five-level eikonal hierarchy (Physical Axis P1–P5).

Level	Name	Tracked	Validity	Key Application
1	Phase	S only	NA < 0.3	Basic imaging, camera lenses
2	Amplitude	S, A	+caustics	Illumination, beam shaping
3	Vector	$S, A, \mathbf{J}$	NA < 0.95	Lithography, microscopy
4	Coherence	$S, A, \mathbf{J}, \mathcal{W}$	Partial coh.	OCT, speckle, LED
5	Nonlinear	All $+\chi^{(n)}$	High I	SPDC, entanglement

Jacobian nesting: $J_1 \subset J_2 \subset J_3 \subset J_4 \subset J_5$. Lower-level tolerances are always optimistic.

Design Rule DR-E.2 — Minimum Sufficient Level

Always start at Level 1. Upgrade *only* when the regret—the merit function difference between levels—exceeds your tolerance budget. Level 1 suffices for $\sim$80% of lens design tasks.

3.1 Level Upgrade Triggers

Table 4: Level upgrade triggers with quantitative thresholds.

Upgrade	Trigger Condition	Typical Case
$1 \to 2$	Uniformity $< 90\%$ or $\lvert J\rvert < 0.1$	Illumination, reflector
$1 \to 3$	NA > 0.3 or birefringence $> \lambda/100$	Microscopy, litho, LC
$1 \to 4$	$\ell_c < 10\times$ feature size	LED, OCT, projection
Any $\to 5$	$I > 1\,\mathrm{MW/cm^2}$	SPDC, SHG, Kerr

Decision rule (DR-E.2b): If NA > 0.3 *and* polarization purity matters: go directly to Level 3. For quantum applications requiring $F > 0.99$: Level 3 or higher is mandatory.

4 The Three-Axis Failure Framework

When the eikonal model fails, diagnose along three orthogonal axes. Check P first (most common), then M, then T.

Table 5: Three-axis failure classification.

Axis	Codes	Question	Symptom
Physical	P1–P5	What are you NOT tracking?	Pred. vs. meas. gap grows with NA
Mathematical	M1–M5	Is the phase well-behaved?	Instability, fringe jumps, aliasing
Topological	T1–T3	Domain simply connected?	OAM non-conservation, mode coupling

Table 6: Mathematical axis failures M1–M5.

Code	Failure	Symptom	Remedy
M1	Discontinuity	Phase jumps	Smooth approx: $\phi_{\mathrm{sm}} = \frac{\phi_{\max}}{2}(1 + \tanh(x/\sigma))$
M2	Singularity	$\nabla S \to \infty$	Regularization
M3	Multi-valued	Multiple S at point	Branch tracking
M4	Undersampling	Aliasing	Adaptive mesh; $N \geq 4 \times (\text{max order})^2$
M5	Ill-conditioning	Matrix near-singular	Preconditioning, float64

Design Rule DR-E.3 — M4 Detection

Compute Strehl at grid sizes N and $2N$. If $\lvert S_{2N} - S_N\rvert/S_N > 0.01$, sampling is insufficient. For aberrations with $\lvert a_j\rvert > 0.2\lambda$, start with $N \geq 256$.

5 The Specification Translation Table

5.1 Tightening Factor

The bridge identity quantifies exactly how much tighter quantum specs must be:

$$\text{Tightening factor} = \frac{\sigma_{W,\text{classical}}}{\sigma_{W,\text{quantum}}} = \frac{0.075\lambda}{0.016\lambda} = 4.7\times \tag{5}$$

For high-fidelity quantum computing ($F > 0.999$), the tightening reaches 14×–45×.

Table 7: Fidelity / Strehl ladder—from consumer imaging to quantum gates.

Application	F/S	σ_W	Tighten	Manufacturing
Consumer imaging	> 0.50	0.132λ	1×	Standard polish
Precision imaging	> 0.80	0.075λ	1×	Precision polish
Diffraction-limited	> 0.95	0.036λ	2×	MRF
Quantum photonics	> 0.99	0.016λ	4.7×	Ion beam figuring
QKD	> 0.99	$\lambda/63$	6×	IBF + active align
Entanglement	> 0.999	$\lambda/200$	14×	Superpolish
Quantum gates	> 0.9999	$\lambda/630$	45×	Extreme tolerance

Key relation: $F = S = e^{-\sigma_\phi^2}$ where $\sigma_\phi = 2\pi\sigma_W/\lambda$.
General formula: To translate from any classical Strehl target S_c to any quantum fidelity target F_q:

$$\text{Tightening factor} = \sqrt{\frac{\ln(1/S_c)}{\ln(1/F_q)}}$$

5.2 Specification Translation Master Table

Table 8: Specification translation: classical → quantum (master table).

Parameter	Classical	Quantum	Factor	Reason
RMS WFE	$\lambda/14$	$\lambda/140$	10×	Fidelity $F > 0.99$
Strehl / Fidelity	> 0.80	> 0.99	—	Error correction
Surface roughness	2 nm	0.2 nm	10×	Scattering loss
Index homogeneity	10^{-5}	10^{-7}	100×	Phase coherence
AR coating	$< 0.5\%$	$< 0.01\%$	50×	Photon loss
Alignment	10 µrad	0.1 µrad	100×	Mode matching
Temp. stability	1 K	10 mK	100×	Phase drift
Detector QE	80%	98%	1.2×	Loss budget

Path-length note: Table 8 values assume typical bench-scale optical paths (0.1–1 m). For shorter paths (e.g., 10 mm PIC), use Eq. (10) to compute system-specific thermal requirements. For meter-scale interferometers, requirements tighten further.

Design Rule DR-E.4 — Quantum Tax

Rule of thumb: quantum = classical tightened by 10–100×. For N-photon states, tolerances tighten by an additional factor of N. When designing quantum wavefront sensors, start with classical specifications and apply the appropriate tightening factors from Table 8.

5.3 Fill-Out Recipe

What To Do First — Row-by-Row Procedure

1. **Define system:** list all surfaces, materials, spacings. Write down λ, NA, field.
2. **Compute** $W(\mathbf{r})$ via ray trace (Level 1). Fit Zernikes: $W = \sum a_n Z_n$.
3. **Compute** $\sigma_W = \sqrt{\sum_{j\geq 4} a_j^2}$. Compute $S = e^{-(2\pi\sigma_W)^2}$.
4. **Check Level:** NA > 0.3? Upgrade (Table 4). Check M4 (DR-E.3).
5. **Bridge:** $\sigma_\phi = 2\pi\sigma_W/\lambda$. Compute $F = S$. Check against spec (Table 7).
6. **If spec fails:** proceed to optimization (§6).

6 The DEE Four-Step Workflow

Algorithm DEE.A — Unified W/MN Workflow

1. **S1: Design [MN].** Find optimal $\mathbf{p}^*$:
```
loss = lambda p:  1 - forward_model(p).strehl
grads = jax.grad(loss)(params)
params = params - lr * grads
```
   Repeat for 200–500 iterations (Adam optimizer).
2. **S2: Verify [W].** Evaluate all metrics at $\mathbf{p}^*$:
```
metrics = forward_model(params_opt)
```
   Check: $S > 0.80$ (classical) or $F > 0.99$ (quantum)?
3. **S3: Tolerance [MN].** Extract manufacturing limits from Hessian:
```
H = jax.hessian(loss)(params_opt)
tol_j = sqrt(2 * delta_F / H[j,j])
```
   Eigendecomposition: $H = V\Lambda V^T$. Identify critical parameter.
4. **S4: Yield [W].** Monte Carlo validation:
   Sample $\mathbf{p} \sim \mathcal{N}(\mathbf{p}^*, \boldsymbol{\sigma}^2)$, $N = 1000$ samples.
   Yield = fraction with $S > S_{\min}$. If yield $< 90\%$: tighten or iterate S1.

Defaults: lr $= 0.01$ (Adam); $\Delta F_{\max} = 0.02$ (2% fidelity drop); $N_{\mathrm{MC}} = 1000$.
Typical timing: S1 $\sim$0.25 s; S2 $\sim$0.3 ms; S3 $\sim$18 ms; S4 $\sim$0.3 s (GPU).

7 Tool Translation Table

Table 9: Rosetta Stone — eikonal level to design action.

Lv	Physics	CODE V	Zemax	DEE Advantage
1	Scalar ray/wave	Ray trace	Sequential	Autodiff gradients
2	+Amplitude	Gaussian beam	POP	Caustic-stable
3	+Polarization	Pol. RT	Jones pupil	Full **J** Jacobian
4	+Coherence	(Limited)	Coh. PSF	Wigner transport
5	+Nonlinear	N/A	N/A	Unique to DEE

Three-layer production workflow: **Exploration** (Python/JAX) $\rightarrow$ **Verification** (CODE V/Zemax/RSoft) $\rightarrow$ **Manufacturing** (GDS/STEP).

Data format translation: `export_codev()`, `export_rsoft()`, `export_gds()`, `export_step()`.

8 Three Limiting Cases

1. **Classical-only** ($F > 0.80$ sufficient): Standard lens design. Use Level 1 DEE with RMS WFE loss. Tolerances at $\lambda/14$. Most commercial imaging.
2. **Quantum-ready** ($F > 0.99$ required): Same DEE code, tighter loss target. Tolerances tighten 4.7–14×. Requires IBF, superpolish. QKD, quantum sensors.
3. **N-photon** ($F > 0.99^{1/N}$ per element): Tolerances tighten by additional factor N. Loss budget dominates. NOON states, quantum lithography.

The DEE's first value is telling you *which case you are in.*

9 Tolerance Matching

The Hessian eigendecomposition reveals which parameters are critical:

$$\sigma_k = \sqrt{\frac{2\,\Delta L_{\max}}{\lambda_k}} \tag{6}$$

where λ_k are eigenvalues of $\mathbf{H}$ and $\Delta L_{\max}$ is the maximum acceptable loss increase.

Table 10: Hessian condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ interpretation.

κ	Interpretation	Action	Quantum Implication
< 10	Well-conditioned	Uniform tolerances	Balanced multi-axis sensor
10–100	Moderate	Selective tighten	Directional sensor; 1–2 dominant axes
100–10^4	Severe	Fix critical param first	Single-axis sensor; high F_Q along one direction
$> 10^4$	Near-singular	Reparameterize	Degenerate; sensing concentrated on one mode

The **Quantum Implication** column reflects the precision–stability duality: a device that is "easy to manufacture" (small H_{ii}) is "hard to use for sensing" (small F_Q), and vice versa.

Design Rule DR-E.5 — Hessian = Tolerance Budget

$\Delta p_j = \sqrt{2\,\Delta F/H_{jj}}$ gives the full tolerance budget in one shot (18 ms on GPU vs. 3.2 hours traditional). The Hessian bridges optimization and manufacturing.

10 The DEE Computational Advantage

Table 11: DEE speedup over traditional finite-difference workflows (36 Zernike terms, 256×256 grid).

Operation	Traditional	DEE (JAX)	Speedup
Forward evaluation	150 ms	0.3 ms	500×
Gradient (all N params)	5.4 s	0.5 ms	10,800×
Hessian ($N \times N$)	194 s	18 ms	10,800×
Full optimization	45 min	0.25 s	10,800×
Tolerance analysis	3.2 hr	0.02 s	**576,000**×
Quantum extension	Separate code	Same code	Unified

Key Insight. The fundamental theorem of reverse-mode autodiff: for $f : \mathbb{R}^N \to \mathbb{R}$, $\mathrm{Cost}(\nabla f) \approx 1.5 \times \mathrm{Cost}(f)$, independent of N. This is why 100 parameters cost the same as 10.

11 System Design Workflow: Six Steps

Optical System Design Process

1. **Define requirements.** $S_{\min}$, $F_{\min}$, field angle, NA, λ, packaging constraints.
2. **Initial design.** Starting point from library/experience. Evaluate $W(\mathbf{r})$ (Walther).
3. **Classify level.** Check NA, polarization, coherence against Table 4. Select minimum sufficient level (DR-E.2).
4. **Apply DEE workflow.** Algorithm DEE.A (§6): Design → Verify → Tolerance → Yield.
5. **Quantum bridge check.** If quantum app: apply bridge identity (Eq. 3). Check F against quantum spec. Apply tightening (Table 8).
6. **Verify and iterate.** Critical parameter may shift after optimization. Recalculate Hessian. Repeat.

12 Condensed Worked Example WE-7.0

Case Study: WE-7.0 — Cooke Triplet Optimization

Setup: Cooke triplet, 6 surfaces; $f = 50\,\text{mm}$, $f/4$, $\lambda = 550\,\text{nm}$, $20°$ half-field. Initial $\sigma_W = 0.155\lambda$ at edge field.

Step 1 — WALTHER (evaluate):
- $\sigma_W = 0.155\lambda$; $S = e^{-(2\pi\times 0.155)^2} = 0.39$
- Spec: $S > 0.80$. **FAIL.**

Step 2 — MATSUI-NARIAI (optimize):
- `loss = lambda p:  rms_wfe(p)`
- `grads = jax.grad(loss)(params)` — 500 iterations, Adam, lr = 0.01
- Result: $\sigma_W \to 0.024\lambda$; $S = 0.978$. **PASS.**

Step 3 — TOLERANCE (Hessian):
- `H = jax.hessian(loss)(params_opt)` — 18 ms on GPU
- Critical parameter: c_4 (curvature of surface 4), $\delta c_4 = \pm 0.016\,\text{mm}^{-1}$
- Monte Carlo yield: 94.2% at $S > 0.80$. **PASS.**

Step 4 — QUANTUM BRIDGE:
- $F = S = 0.978$; classical spec met.
- Quantum spec ($F > 0.99$): **FAIL.** Need $\sigma_W < 0.016\lambda$.
- Required: tighten c_4 to $\pm 0.0016\,\text{mm}^{-1}$ ($10\times$); quantum yield $< 5\%$ with current process.

Step 5 — QUANTUM RE-OPTIMIZATION *(new in v4.2):*
- Replace loss: `loss_q = lambda p:  1 - fidelity(p)` with target $F > 0.99$
- Re-run S1: 800 iterations, Adam, lr = 0.005 (smaller step for tighter target)
- Result: $\sigma_W \to 0.014\lambda$; $F = 0.993$. **PASS (quantum).**
- Re-run S3: $\delta c_4 = \pm 0.0012\,\text{mm}^{-1}$ (requires IBF process)
- Re-run S4: Quantum yield with IBF process: 67%. **MARGINAL.** Requires further tolerance rebalancing or process upgrade.

Narrative closure: Same forward model, same DEE code, same 7-line workflow. Classical → quantum required only changing the loss function target and re-running. The 576,000× speedup makes this interactive—the entire 5-step sequence completes in under 2 seconds on GPU.

13 Hessian–QFI Connection

Quantum Extension — Tolerancing = Quantum Sensing

The classical Hessian from DEE tolerance analysis relates directly to the Quantum Fisher Information:

$$\boxed{F_Q = 4\,\mathbf{H}_{\text{fidelity}}} \tag{7}$$

This means: tight fabrication tolerances (small $\Delta\kappa$) correspond to high quantum sensing capability. A device designed for precision classical applications is automatically well-suited for quantum sensing.

Quantum Cramér–Rao bound: $\delta\theta_{\min} = 1/\sqrt{N \cdot F_Q(\theta)}$.

Table 12: Quantum sensing key limits.

Limit	Equation	Scaling
Standard Quantum Limit	$\delta\phi_{\text{SQL}} = 1/\sqrt{N}$	Shot noise
Heisenberg Limit	$\delta\phi_{\text{HL}} = 1/N$	Ultimate
Squeezed state	$\delta\phi = e^{-r}/\sqrt{N}$	Practical
Loss degradation	$r_{\text{eff}} = r - \frac{1}{2}\ln(1/\eta)$	Efficiency budget
Quantum advantage	$Q = e^{r_{\text{eff}}}$	Enhancement
N-photon phase	$\phi_N = N \cdot \phi_1$	Multiplication
Loss fidelity (11% rule)	$F = e^{-0.115\,\alpha L}$	Per-cm loss

14 Eigenmode-Eikonal Identity (Waveguides)

For coupled waveguide arrays with coupling matrix $\mathbf{C}$:

$$\phi_k = \beta_k z, \quad \beta_k = \beta_0 + 2\kappa\cos\left(\frac{k\pi}{N+1}\right) \tag{8}$$

Key Insight. Eigenvalues of the coupling matrix *are* the eikonals for the coupled system. This unifies classical coupler design and quantum photonic gates. The same DEE-Waveguide module optimizes both.

HOM dip: $P_{\text{coinc}} = (1 - V)/2$. Quantum walk spreading: $\sigma_{\text{qu}} \propto z$ (ballistic) vs. $\sigma_{\text{cl}} \propto \sqrt{N}$ (diffusion).

Design Rule DR-E.6 — Coupling Uniformity

For quantum walks, coupling uniformity must satisfy $\sigma_\kappa/\kappa_0 < 10\%$ to avoid disorder-induced localization. This is the #1 fabrication challenge for waveguide quantum photonics.

15 N-Photon Phase Multiplication

$$\phi_N = N \times \frac{2\pi}{\lambda} W_{\text{eikonal}} \tag{9}$$

- Sensitivity scales as $N\times$ (super-resolution).
- Tolerances tighten by $N\times$ (manufacturing harder).
- Effective enhancement: $\eta^N \cdot N$ (loss is exponential).
- Quantum advantage: $Q = \eta^N \cdot N/\sqrt{N} = \eta^N \sqrt{N}$.

$Q > 1$ requires $\eta > N^{-1/(2N)}$. For $N = 4$: $\eta > 0.84$ (84% efficiency per element).

Design Rule DR-E.7 — N-Photon Feasibility

Before committing to N-photon imaging: compute $Q = \eta^N \sqrt{N}$. If $Q < 1$, classical beats quantum. **Loss is the gatekeeper, not resolution.**

16 Thermal–Optical Bridge

Trilogy Connection — Thermal → Optical Error (Book 3)

$$\text{WFE} \approx \frac{(dn/dT) \cdot \Delta T \cdot L}{\lambda} \quad \text{[waves]} \tag{10}$$

If WFE $> \lambda/10$, tighten thermal design. Every thermal path through a refractive element should carry a WFE estimate. Connects to the Mode-Path Matrix in Book 3 (*Thermal Photonics*).

For quantum systems ($F > 0.99$): WFE $< \lambda/63$, requiring $\Delta T < \lambda^2/(63 \cdot L \cdot dn/dT)$.
At $\lambda = 1550$ nm, $L = 10$ mm, $dn/dT = 10^{-5}$/K: $\Delta T < 0.38$ K.

Path-length scaling (v4.2 clarification): Table 8 quotes $\Delta T < 10$ mK for quantum applications. This assumes meter-scale interferometric paths ($L \sim 1$ m). Equation (10) is the system-specific formula; always compute ΔT for *your* path length. Thermal stability is typically the **most expensive** tolerance to achieve in quantum photonics—budget 30–50% of total WFE to thermal drift.

17 Classical–Quantum Failure Coupling

Classical failures amplify in quantum systems. This table explains *why* the 5–50× tightening factors in Table 8 arise mechanistically.

Table 13: Classical–quantum failure coupling with amplification factors.

Classical Failure	Triggers Quantum	Amplification	Mechanism
P1 (High-NA)	Q4 (Indistinguishability)	3–5×	Vector depolarization reduces photon purity
P2 (Diffraction)	Q5 (Temporal mismatch)	2–3×	Broadened path-length spread
M1 (Discontinuity)	Q6 (Phase drift)	5–10×	Phase noise amplified by edge scattering
M2 (Caustic)	Q8 (Spectral leak)	10–20×	Energy redistribution corrupts spectral modes
T2 (Mode coupling)	Q2 (Loss cascade)	2–5×	Mode leakage becomes photon loss

Quantum failure codes (from Chapter 12): Q1 squeezing degradation, Q2 photon loss cascade, Q3 NOON fragility, Q4 indistinguishability loss, Q5 temporal mismatch, Q6 phase drift, Q7 polarization decoherence, Q8 spectral entanglement leak.

Design Rule (implicit): For quantum applications, check classical failure modes *first*. If any classical failure triggers, the quantum system will almost certainly fail. Fix classical issues before addressing quantum-specific concerns.

18 Common Eikonal Bridge Failure Modes

Top Eight Practitioner Mistakes

1. **Using Level 1 beyond NA > 0.3 (P3).** Scalar eikonal ignores vector depolarization. An NA $= 0.85$ objective with predicted $S = 0.89$ may actually give $S = 0.72$. *Prevention:* check NA against Table 4.
2. **Optimizing Strehl directly.** Strehl loss is non-convex and has local minima. *Prevention:* optimize RMS WFE (strongly convex) first; refine in Strehl space only for final tuning.
3. **Ignoring M4 (sampling violation).** 64×64 grid with $a_{11} = 0.5\lambda$ gives $S = 0.82$; 256×256 gives $S = 0.67$. *Prevention:* always run DR-E.3 convergence check.
4. **Applying classical tolerances to quantum systems.** $\lambda/14$ WFE is fine for $S > 0.8$ but gives $F = 0.8$—catastrophic for quantum ($F > 0.99$ needed). *Prevention:* apply Table 8 tightening factors.
5. **Trusting Hessian at large perturbations.** Quadratic approximation fails for $|\Delta p| > 2\sigma$. *Prevention:* always validate S4 with Monte Carlo (Algorithm DEE.A).
6. **Gradient vanishing or explosion.** $\|\nabla\mathcal{L}\| < 10^{-12}$ or $\|\nabla\mathcal{L}\| > 10^{6}$. *Prevention:* use float64; rescale parameters; add regularization.
7. **Model-reality gap (P1–P5).** Fabricated device doesn't match simulation. *Prevention:* upgrade eikonal level per Table 4; check all three axes (§4).
8. **Not re-checking after optimization (DR-E.8).** Critical parameter may shift after one parameter is tightened. *Prevention:* rebuild Hessian after every design change; repeat Algorithm DEE.A S3–S4.

19 Navigation Routing Guide

- Level 1 (NA < 0.3, scalar) $\to$ Ch. 1–3 + Ch. 7. Standard lens design.
- Level 2–5 (failure detected) $\to$ Ch. 4. Three-axis diagnosis.
- Waveguide/PIC design $\to$ Ch. 5. Eigenmode-eikonal.
- Global optimization needed $\to$ Ch. 6. SA, DE, QUBO, no-regret.
- DEE implementation $\to$ Ch. 7. JAX pipeline, Hessian, tolerance.
- Duality reference $\to$ Ch. 8. Master W/MN table.
- Quantum wavefront sensing $\to$ Ch. 9. SQL $\to$ Heisenberg.
- Quantum walks / HOM $\to$ Ch. 10. Correlation functions.
- N-photon imaging $\to$ Ch. 11. Phase multiplication.
- Production / yield $\to$ Ch. 12. Monte Carlo, three-layer workflow.
- Thermal coupling $\to$ Book 3 (*Thermal Photonics*), Eq. (10).

20 Master W/MN Reference Table

Table 14: Master Walther / Matsui-Nariai reference across all chapters.

Domain	Walther (Forward)	Matsui-Nariai (Inverse)	Ch.
Ray tracing	Params $\to$ rays?	Target $\to W$?	1,2
Aberration	Zernike $\to$ PSF?	PSF $\to$ Zernike budget?	3
Level select	Design $\to$ failure axis?	Constraint $\to$ min level?	4
Waveguide	$C \to P(z)$?	$P_{\text{tgt}} \to C$?	5
Optimization	Evaluate M?	Minimize M?	6
DEE gradients	`forward_model()`	`jax.grad(loss)`	7
Duality	All analysis	All design	8
Quantum WFS	System $\to \delta\phi$?	$\delta\phi_{\text{tgt}} \to$ system?	9
Quantum walk	Array $\to$ distribution?	Distribution $\to$ array?	10
N-photon	N, loss $\to$ resolution?	Resolution $\to N$, loss?	11
Production	Process $\to$ yield?	Yield $\to$ process?	12

21 Loss Function Selection Guide

Table 15: Loss functions in DEE — selection guide.

Loss Function	Formula	Convexity	Use Case
RMS WFE	$\sqrt{\sum a_j^2}$	Strongly convex	General (use first)
Strehl	$1 - S$	Non-convex	Direct metric (refine)
MTF at freq f	$1 - \mathrm{MTF}(f)$	Non-convex	Resolution
Encircled energy	$1 - \mathrm{EE}(r)$	Non-convex	Concentration
Quantum fidelity	$1 - \lvert\langle\psi\vert\psi_{\mathrm{tgt}}\rangle\rvert^2$	Non-convex	Quantum apps

Design Rule DR-E.9 — Loss Function Strategy

Start with RMS WFE (strongly convex, guaranteed global minimum). Switch to direct metric (Strehl, fidelity) only for final refinement. Never start with a non-convex loss for initial design.

This is not a heuristic—it is a theorem. The RMS WFE loss has a unique global minimum in Zernike coefficient space; the Strehl loss does not. Practitioners can trust monotonic convergence during the convex phase and should expect non-monotonic behavior only during the refinement phase.

22 JAX Quick Reference and Minimal Starter

Table 16: Essential JAX patterns for optical design.

Operation	JAX Call
Forward eval	`f(params)`
Gradient	`jax.grad(f)(params)`
Hessian	`jax.hessian(f)(params)`
Jacobian	`jax.jacfwd(f)(params)`
Vectorize (MC)	`jax.vmap(f)(batch_params)`
JIT compile	`jax.jit(f)`
GPU/TPU check	`jax.devices()`

Minimal JAX Starter — the complete DEE pattern in 12 lines:

```
import jax
import jax.numpy as jnp

def forward_model(params):
    """Zernike -> RMS WFE (WALTHER)."""
    return jnp.sqrt(jnp.sum(params[3:]**2))

loss = lambda p: forward_model(p)     # Classical
loss_q = lambda p: 1.0 - jnp.exp(    # Quantum
    -(2*jnp.pi*forward_model(p))**2)

grads = jax.grad(loss)(params)        # MATSUI-NARIAI
H = jax.hessian(loss)(params)         # Tolerance
```

Write `forward_model()` once. Classical: `loss = rms_wfe(p)`. Quantum: `loss_q = 1 - fidelity(p)`. Gradients, Hessians, and tolerances follow identically from the same forward model.

23 Complete Design Rules Summary

Table 17: Eikonal Bridge design rules (DR-E series) — complete set of 11.

Rule	Key	Description	Eq.
DR-E.1	W/MN classify	Every problem is Walther or Matsui-Nariai.	—
DR-E.2	Min. level	Start Level 1; upgrade only when regret exceeds budget.	(2)
DR-E.2b	Level 3 gate	NA > 0.3 AND polarization $\to$ Level 3 directly.	—
DR-E.3	M4 detect	$\lvert S_{2N} - S_N\rvert / S_N > 0.01 \to$ increase grid.	—
DR-E.4	Quantum tax	Quantum = classical × 10–100 tightening.	(5)
DR-E.5	Hessian = tol	$\Delta p_j = \sqrt{2\Delta F / H_{jj}}$; one shot, 18 ms.	(6)
DR-E.6	Coupling unif.	$\sigma_\kappa / \kappa_0 < 10\%$ for quantum walks.	(8)
DR-E.7	N-photon gate	$Q = \eta^N \sqrt{N} > 1$ or classical wins.	(9)
DR-E.8	Re-check	Rebuild Hessian after every design change.	—
DR-E.9	Loss strategy	RMS WFE first; metric loss for refinement only.	—
DR-E.10	Bridge always	$F = S$; check quantum spec for every design.	(3)
DR-E.11	Unity rule	There is no separate quantum design tool—only one tool with two loss functions.	(BI)

Design Rule DR-E.11 — The Unity Rule (new in v4.2)

There is no "quantum optical design tool" separate from your classical one. There is only **one tool with two loss functions**. If you can do classical lens design with DEE, you can already do quantum photonic design. The bridge identity (BI) is the *only* equation you need to cross over.

Table 18: DR numbering legacy map.

Prefix	Ch. Origin	Topic	Notes
DR-E.n	QRG v4.2	Eikonal Bridge	This document
DR-P.n	Ch. 4	Physical axis (P1–P5)	Level hierarchy
DR-M.n	Ch. 4	Mathematical axis (M1–M5)	Regularity failures
DR-T.n	Ch. 4	Topological axis (T1–T3)	Structural obstructions
DR-7.n	Ch. 7	DEE implementation	JAX patterns
DR-8.n	Ch. 8	W/MN duality	Workflow steps
DR-Q.n	Ch. 9–11	Quantum extensions	Sensing, walks, N-photon
DR-12.n	Ch. 12	Production	Yield, cost, fab

Table 19: Eikonal Bridge applications across all 12 chapters.

Ch.	Bridge Application
1	Home of bridge identity $\phi = 2\pi W/\lambda$. Singlet practical example.
2	Hamilton's 4 characteristic functions; point/angle eikonals; propagator connection.
3	Zernike $\to$ PSF $\to$ Strehl pipeline; $F = S$ demonstration; M4 sampling failure.
4	Five-level hierarchy; three-axis failure framework; level upgrade decision tree.
5	Eigenmode-eikonal for waveguides; CPO design; quantum photonic gates.
6	Quantum-inspired optimization; QUBO; no-regret strategy for quantum adoption.
7	DEE architecture; JAX autodiff; Hessian tolerance; 10,800× speedup.
8	W/MN duality unified; master reference table; four-step workflow.
9	Quantum wavefront sensing; SQL $\to$ squeezed $\to$ Heisenberg; spec tightening.
10	Quantum walks in waveguide arrays; HOM; Hessian-QFI connection $F_Q = 4H$.
11	N-photon phase multiplication; quantum advantage $Q = \eta^N \sqrt{N}$.
12	Lab to fab; three-layer workflow; Monte Carlo yield; failure mode catalog.
A	Math reference: Zernike polynomials, Jones matrices, operator algebra.
B	Code repository guide: DEE API, installation, quick start.
C	Specification translation tables: complete classical $\leftrightarrow$ quantum mapping.
D	Quantum optics primer for optical engineers.

Table 20: Key references and their bridge connections.

Reference	Bridge Connection
Hamilton (1828)	Optical-mechanical analogy; foundation of eikonal theory.
Bruns (1895)	Named the "eikonal"; $\lvert\nabla W\rvert^2 = n^2$.
Walther (1995), *The Ray and Wave Theory of Lenses*	Forward analysis paradigm; ray-optical phase space; foundation of the W-half of W/MN duality.
Matsui & Nariai (1993), *Fund. of Practical Aberration Theory*	Inverse design paradigm; structural aberration formulas; starting point generation; foundation of the MN-half.
Born & Wolf (1999)	Classical diffraction and aberration theory; Zernike formalism.
Goodman, *Intro. Fourier Optics*	FFT-based PSF; coherence theory (Level 4).
Bradbury et al., JAX (2018)	Autodiff framework; reverse-mode gradients; GPU acceleration.
Shannon, *Art & Science of Optical Design*	Practical lens design; optimization wisdom.
Mandel & Wolf (1995)	Quantum coherence theory; bridge to Level 4/5.
Maréchal (1947)	$S \approx e^{-(2\pi\sigma_W)^2}$; the approximation that makes $F = S$ useful.
Giovannetti et al. (2004, 2011)	Quantum metrology; SQL and Heisenberg limits.
Caves (1981)	Quantum-mechanical noise in interferometers; squeezing.

Eikonal Bridge Quick Reference v4.2

Companion to *"The Eikonal Bridge: From Classical Lens Design to Quantum Photonics"*

Book 1 of the Open-Source Engineering Optics Trilogy

Cross-references: *Quantum Field Imaging* (field correlations in detection);
Thermal Photonics (thermal $\to$ optical error, MPM–WFE bridge)

#OpticalEngineering #QuantumPhotonics #LensDesign #SPIE #DifferentiableComputing #JAX

22 Sections · 21 Tables · 11 Design Rules · Top 8 Failure Modes · 1 Bridge Identity

One equation. Two domains. Unified code.